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A Variational Approach to the Theory of the Effective Magnetic Permeability of Multiphase Materials*

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Variational theorems are established and applied to the derivation of bounds for the effective magnetic permeability of macroscopically homogeneous and isotropic multiphase materials. For reasons of mathematical analogy the results are also valid for the dielectric constant, electric conductivity, heat conductivity, and diffusivity of such materials. For the case of two-phase materials, the bounds derived are the most restrictive ones that can be given in terms of the phase permeabilities and volume fractions. Comparison of present theoretical results with existing experimental data shows good agreement.

I. INTRODUCTION

The present work[1] is concerned with the problem of the theoretical determination of the effective magnetic permeability of macroscopically homogeneous and isotropic multiphase materials, in terms of the volume fractions and permeabilities of the constituting phases. For reasons of mathematical analogy the results obtained also hold for the dielectric constant, electric conductivity, heat conductivity, and diffusivity of such materials.

The problem is one of long standing and has been treated in many papers on the basis of various simplifying assumptions. A large number of different results has thus been obtained. A review of previous work is, however, not within the scope of this paper. Such reviews may be found, for example, in references 2–6.

A fundamental question is whether the magnetic permeability of a multiphase material is completely determined by the volume fractions and magnetic permeabilities of the phases. For a two-phase material Brown,[7] who studied the dielectric case, has concluded that such information as mentioned above is not sufficient and that the statistics of phase distribution influences the result. This has also been supported by experiments of DeLoor.[3] Therefore, instead of attempting to derive an expression for the effective permeability, the purpose of this paper is the derivation of upper and lower bounds for this quantity. These bounds are derived by use of some variational theorems which are established below. It is furthermore shown that in the case of a two-phase material these bounds are the most restrictive ones that can be obtained in terms of phase volume fractions and permeabilities. Finally, the theoretical results are compared with experimental measurements and good agreement is found.

II. VARIATIONAL PRINCIPLES[8]

Let **B** and **H** denote the magnetic induction and field intensity, respectively, obeying

$$\operatorname{div}\mathbf{B}=0, \tag{2.1}$$

$$\operatorname{curl}\mathbf{H}=0. \tag{2.2}$$

Equation (2.2) may also be written as

$$\mathbf{H}=-\operatorname{grad}\psi. \tag{2.3}$$

Consider a homogeneous body of volume V and surface S consisting of an isotropic material of permeability μ_0. Let $\mathbf{B}_0$ and $\mathbf{H}_0$ denote the magnetic induction and field intensity, respectively, when a surface potential

$$\psi(S)=\psi_0(S) \tag{2.4}$$

is prescribed. The vector fields $\mathbf{B}_0$ and $\mathbf{H}_0$ must satisfy (2.1) and (2.2), respectively, and are related by

$$\mathbf{B}_0=\mu_0\mathbf{H}_0. \tag{2.5}$$

Let part or whole of the body now be changed to a material of different permeability μ, which may vary in space, without however changing the surface potential.

Let the generalized magnetization **T** now be defined by

$$\mathbf{T}=\mathbf{B}-\mu_0\mathbf{H}. \tag{2.6}$$

Define also

$$\psi'=\psi-\psi_0, \tag{2.7}$$

$$\mathbf{H}'=\mathbf{H}-\mathbf{H}_0, \tag{2.8}$$

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[1] Some of the results of the present study were briefly reported previously. See Z. Hashin and S. Shtrikman, J. Franklin Institute **271**, 423 (1961).

[2] C. J. F. Böttcher, *Theory of Electric Polarization* (Elsevier Publishing Company, Houston, Texas, 1952), pp. 415–420.

[3] G. P. DeLoor, "Dielectric Properties of Heterogenous Mixtures," Thesis, University of Leiden, Leiden (1956).

[4] J. A. Reynolds and G. M. Hough, Proc. Phys. Soc. (London) **B70**, 796 (1957).

[5] C. Herring, J. Appl. Phys. **31**, 1939 (1960).

[6] W. Woodside and J. H. Messner, J. Appl. Phys. **32**, 1688 (1961).

[7] W. F. Brown, Jr., J. Chem. Phys. **23**, 1514 (1955).

[8] These principles are an extension of a variational theorem formulated by W. F. Brown, Jr., in a private communication.

Then the volume integral

$$U_T=\frac{1}{8\pi}\int_{(V)}\left(\mu_0\mathbf{H}_0{}^2-\frac{\mathbf{T}^2}{\mu-\mu_0}+2\mathbf{T}\cdot\mathbf{H}_0+\mathbf{T}\cdot\mathbf{H}'\right)dv, \quad (2.9)$$

subject to the subsidiary condition

$$\mu_0\,\mathrm{div}\mathbf{H}'+\mathrm{div}\mathbf{T}=0 \quad (2.10)$$

and the boundary condition

$$\psi'(S)=0, \quad (2.11)$$

is stationary for

$$\mathbf{T}=(\mu-\mu_0)\mathbf{H}. \quad (2.12)$$

This is proved in the Appendix. It is furthermore shown in the Appendix that the stationary value of U_T, denoted by $U_T{}^S$, is an absolute maximum when

$$\mu_0<\mu,$$

in which case

$$U_T<U_T{}^S, \quad (2.13a)$$

and an absolute minimum when

$$\mu_0>\mu,$$

in which case

$$U_T>U_T{}^S. \quad (2.13b)$$

Note that (2.12) is equivalent, because of (2.6), to

$$B=\mu H,$$

and that (2.11) when substituted into (2.7) ensures that (2.4) holds. Thus the fields $\mathbf{T}$, $\mathbf{H}$, and $\mathbf{B}$, for which (2.9) is stationary, are the solution of the boundary value problem considered.

The stationary value of U_T is equal to the magnetostatic energy stored in V. To show this $U_T{}^S$ is calculated by substituting (2.12) into (2.9) and using (2.6) and (2.8), which yields

$$U_T{}^S=\frac{1}{8\pi}\int_{(V)}\{\mathbf{H}\cdot\mathbf{B}-(\mathbf{B}_0+\mathbf{B})\cdot\mathbf{H}'\}dv. \quad (2.14)$$

The second term on the right-hand side of (2.14) integrates to zero because it follows from the divergence theorem that

$$\int_{(V)}\mathbf{B}\cdot\mathrm{grad}\psi'dv=0, \quad (2.15a)$$

if

$$\psi'(S)=0 \quad (2.15b)$$

and

$$\mathrm{div}\mathbf{B}=0. \quad (2.15c)$$

Accordingly, (2.14) reduces to

$$U_T{}^S=\frac{1}{8\pi}\int_{(V)}\mathbf{H}\cdot\mathbf{B}dv. \quad (2.16)$$

The special limiting cases $\mu_0\to 0$ and $\mu_0\to\infty$ are of interest. For

$$\mu_0\to 0, \quad (2.17a)$$

Eq. (2.6) reduces to

$$\mathbf{T}=\mathbf{B}. \quad (2.17b)$$

Substituting (2.17a) and (2.17b) in (2.9) and using (2.15) it is found that

$$U_T=\frac{1}{8\pi}\int_{(V)}[-(\mathbf{B}^2/\mu)-2\mathbf{B}\cdot\mathbf{H}_0]dv, \quad (2.17c)$$

which on using the divergence theorem is equal to

$$-U_T=\frac{1}{8\pi}\int_{(V)}(\mathbf{B}^2/\mu)dv+\int_{(S)}\psi_0(S)\mathbf{B}\cdot d\mathbf{S}. \quad (2.17d)$$

Condition (2.10) reduces to (2.1) and (2.11) is discarded.

In the case

$$\mu_0\to\infty \quad (2.18a)$$

substitute (2.6) and (2.8) into (2.9), divide by μ_0, and expand with respect to $\mu_0{}^{-1}$ to first order to obtain

$$\frac{U_T}{\mu_0}=\frac{1}{8\pi}\int_{(V)}\left[\frac{-\mathbf{B}}{\mu_0}(\mathbf{H}-\mathbf{H}_0)+\mathbf{H}_0\cdot(\mathbf{H}-\mathbf{H}_0)+\frac{\mu}{\mu_0}\mathbf{H}^2\right]dv, \quad (2.18b)$$

which because of (2.15) reduces to

$$U_T=\frac{1}{8\pi}\int_{(V)}\mu\mathbf{H}^2dv. \quad (2.18c)$$

Equation (2.10) is discarded, but $\mathbf{H}$ should of course obey (2.3) and (2.4).

The variational formulation given above applies to the case of prescribed surface potential. It should be noted that an analogous formulation can be given when, instead, the normal component B_{0n} of the surface induction is prescribed.

For this purpose the vectors $\mathbf{R}$ defined by

$$\mathbf{R}=\mathbf{H}-\mathbf{B}/\mu_0 \quad (2.19)$$

and $\mathbf{B}'$ defined by

$$\mathbf{B}'=\mathbf{B}-\mathbf{B}_0 \quad (2.20)$$

are introduced.

Then the integral

$$U_R=\frac{1}{8\pi}\int_{(V)}\left\{\frac{\mathbf{B}_0{}^2}{\mu_0}-\frac{\mathbf{R}^2}{\mu^{-1}-\mu_0{}^{-1}}+2\mathbf{R}\cdot\mathbf{B}_0+\mathbf{R}\cdot\mathbf{B}'\right\}dv, \quad (2.21)$$

subject to the subsidiary condition

$$\mu_0{}^{-1}\,\mathrm{curl}\mathbf{B}'+\mathrm{curl}\mathbf{R}=0 \quad (2.22)$$

and the boundary condition

$$B'_n=0, \tag{2.23}$$

is stationary for

$$\mathbf{R}=(\mu^{-1}-\mu_0^{-1})\mathbf{B}. \tag{2.24}$$

The stationary value $U_R{}^S$ of U_R is an absolute maximum for $\mu_0>\mu$ and an absolute minimum for $\mu_0<\mu$. The proof of the last two principles is similar to that, given in the Appendix, for the first two.

As before, the stationary value of U_R is equal to the magnetostatic energy stored in the body. Thus

$$U_R{}^S=\frac{1}{8\pi}\int_{(V)}\mathbf{H}\cdot\mathbf{B}dv. \tag{2.25}$$

The proof is analogous to the one given above.

Also in analogy to (2.17) and (2.18) it is found that for $1/\mu_0\to 0$,

$$-U_R=\frac{1}{8\pi}\int_{(V)}\mu\mathbf{H}^2dv+\int_{(S)}\psi B_n(S)dS, \tag{2.26}$$

and for $1/\mu_0\to\infty$,

$$U_R=\frac{1}{8\pi}\int_{(V)}(\mathbf{B}^2/\mu)dv. \tag{2.27}$$

III. BOUNDS FOR THE EFFECTIVE PERMEABILITY OF MULTIPHASE MATERIALS

The effective permeability μ^* of a macroscopically homogeneous and isotropic multiphase material is defined as follows: Apply on the surface of the body considered a potential $\psi_0(S)$ which is derived from a homogeneous field $\mathbf{H}_0$ and calculate the magnetostatic energy U stored in the body. Replace the multiphase material from which the body is composed by a homogeneous and isotropic material, without changing the surface potential. The value of the permeability of the new material, in which the energy stored is equal to that stored in the multiphase material, is defined as the effective permeability of the multiphase material considered. Formally this means[9]

$$\mu^*=U/V\mathbf{H}_0^2. \tag{3.1}$$

Accordingly, to calculate μ^* accurately it is necessary to solve (2.1) and (2.2) subject to the boundary condition on ψ when

$$\mathbf{B}=\mu\mathbf{H}, \tag{3.2}$$

where μ varies erratically in space. This is certainly not easy to carry out, if at all possible. A simpler approach, which will be followed here, consists of the establishment of bounds on μ^* by use of the variational principle (2.9)–(2.13).

[9] Note that because of (2.15) the definition (3.2) is also equivalent to defining the effective permeability as the ratio of the average induction to the average field intensity.

Let the multiphase material consist of m phases. To identify the different phases a running index t will be used:

$$t=1,2,3\cdots m. \tag{3.3a}$$

Accordingly, the permeability of the phase t will be denoted by μ_t; its volume fraction will be denoted by v_t. Therefore,

$$\sum_{t=1}^{m} v_t=1. \tag{3.3b}$$

For convenience it will be assumed that μ_t increases monotonously with t. In order to calculate U_T a trial magnetization field $\mathbf{T}$ has to be chosen. It turns out that a convenient and useful choice is one where $\mathbf{T}$ changes only from phase to phase but is homogeneous within a single phase. With this choice the integration with respect to V can be immediately replaced, for the first three terms of (2.9), by summation with respect to t, so that

$$\frac{8\pi U_T}{V}=\sum_{t=1}^{m}\left\{\mu_0\mathbf{H}_0^2-\frac{\mathbf{T}_t^2}{\mu_t-\mu_0}+2\mathbf{T}_t\cdot\mathbf{H}_0\right\}v_t+\frac{1}{V}\int_{(V)}\mathbf{T}\cdot\mathbf{H}'dv. \tag{3.4}$$

To integrate the last term on the right-hand side of (3.4) a technique due to Néel[10] is used. A cube of reference whose volume L^3 is small compared to the body but large compared to the inhomogeneities in the material is chosen, within which $\mathbf{T}$ is expanded in a Fourier series

$$\mathbf{T}=\langle\mathbf{T}\rangle_{\mathrm{av}}+\sum{}'\,\mathbf{T}_{\mathbf{k}}\exp[(2\pi i/L)\mathbf{k}\cdot\mathbf{r}]. \tag{3.5}$$

Here $\langle\mathbf{T}\rangle_{\mathrm{av}}$ is the average of $\mathbf{T}$, $\mathbf{k}$ is a vector whose components k_x, k_y, k_z are integers and in the summation they vary from $-\infty$ to $+\infty$ excluding only $\mathbf{k}=0$. Also because $\mathbf{T}$ is real

$$\mathbf{T}_{\mathbf{k}}=\mathbf{T}_{-\mathbf{k}}^*. \tag{3.6}$$

The potential ψ' is expanded into

$$\psi'=\sum\psi_{\mathbf{k}}'\exp[(2\pi i/L)\mathbf{k}\cdot\mathbf{r}]. \tag{3.7}$$

The magnetization $\mathbf{T}$ and the potential ψ are related by (2.10), i.e.,

$$\psi_{\mathbf{k}}'=\frac{1}{\mu_0}\frac{L}{2\pi i}\frac{\mathbf{k}\cdot\mathbf{T}_{\mathbf{k}}}{\mathbf{k}^2}. \tag{3.8}$$

From (3.5), (3.7), (3.8), and (3.6) it is found that

$$U'=\frac{1}{V}\int_{(V)}(\mathbf{T}\cdot\mathbf{H}')dv=-\frac{1}{\mu_0}\sum{}'(\mathbf{k}\cdot\mathbf{T}_{\mathbf{k}})(\mathbf{k}\cdot\mathbf{T}_{\mathbf{k}})^*\frac{1}{\mathbf{k}^2}, \tag{3.9}$$

[10] L. Néel, J. phys. radium 9, 184 (1948).

where the integration is carried out over the cube of reference. Note that the last term in the right-hand side of (3.4) should have been computed for the whole body. Because of the assumed macroscopic homogeneity of the multiphase material this quantity will be the same if computed for a large enough reference cube.

Since the material studied is macroscopically isotropic, $\mathbf{T}_\mathbf{k}$ depends on the magnitude k of $\mathbf{k}$ only. Replacing the summation by integration, (3.9) simplifies to

$$\frac{1}{V}\int_{(V)}(\mathbf{T}\cdot\mathbf{H}')dv=-\frac{4\pi}{3\mu_0}\int_0^\infty \mathbf{T}_k\cdot\mathbf{T}_k{}^*k^2dk. \quad (3.10)$$

It should be noted that (3.7) does not satisfy (2.11). However, the error in (3.10) due to this can be neglected, as the cube of reference has been assumed to be large compared to the inhomogeneity scale. This means that $\mathbf{T}_k$, and therefore $\psi_\mathbf{k}$, will be appreciable only for large $\mathbf{k}$ so that the error will be a surface effect.

To evaluate the right-hand side of (3.10) the average of $\mathbf{T}^2$ obtained from squaring (3.5) is calculated. Using a procedure similar to that by which U was calculated it is found that

$$\langle\mathbf{T}^2\rangle_{\text{av}}=V^{-1}\int_{(V)}\mathbf{T}^2dv=\langle\mathbf{T}\rangle^2{}_{\text{av}}+4\pi\int_0^\infty\mathbf{T}_k\cdot\mathbf{T}_k{}^*k^2dk. \quad (3.11)$$

Using the relations

$$\langle\mathbf{T}\rangle_{\text{av}}=\sum_{t=1}^m\mathbf{T}_tv_t, \quad (3.12)$$

$$\langle\mathbf{T}^2\rangle_{\text{av}}=\sum_{t=1}^m\mathbf{T}_t{}^2v_t, \quad (3.13)$$

and (3.10) and (3.11), (3.4) can be written as

$$\frac{8\pi U_T}{V}=\mu_0\mathbf{H}_0{}^2+\sum_{t=1}^m\left(-\frac{\mathbf{T}_t{}^2}{\mu_t-\mu_0}+2\mathbf{T}_t\mathbf{H}_0\right)v_t -\frac{1}{3\mu_0}\{\sum_{t=1}^m\mathbf{T}_t{}^2v_t-(\sum_{t=1}^m\mathbf{T}_tv_t)^2\}. \quad (3.14)$$

[*Note added in proof*. This result can be obtained in a more rigorous way by the use of Fourier transforms.] Combining (2.13), (2.16), (3.1), (3.14), bounds on μ^* are obtained. To find the best bounds for the present magnetization field, (3.14) has to be maximized when (2.13a) applies, and minimized when (2.13b) applies. Differentiating (3.14) with respect to the $\mathbf{T}_t$, the extremum conditions are found to be

$$\mathbf{T}_t=\frac{\mathbf{H}_0+\langle\mathbf{T}\rangle_{\text{av}}/3\mu_0}{(\mu_t-\mu_0)^{-1}+(3\mu_0)^{-1}}, \quad (3.15)$$

which turns out to be a maximum condition for $\mu_0<\mu_t$ and a minimum condition for $\mu_0>\mu_t$.

Multiplying (3.15) by v_t, summing over t, and solving for $\langle\mathbf{T}\rangle_{\text{av}}$, it is found that

$$\langle\mathbf{T}\rangle_{\text{av}}=A\mathbf{H}_0/(1-\alpha A), \quad (3.16)$$

where

$$\alpha=(3\mu_0)^{-1} \quad (3.17a)$$

and

$$A=\sum_{t=1}^m v_t/[(\mu_t-\mu_0)^{-1}+(3\mu_0)^{-1}]. \quad (3.17b)$$

Substituting (3.15) and (3.16) into (3.14), using (3.17), (2.13), (2.16), and (3.2), it is found that

$$\mu^*\lessgtr\mu_0+\frac{A}{1-\alpha A}\quad\text{for}\quad\mu_t\gtrless\mu_0. \quad (3.18)$$

The, as yet unspecified, permeability μ_0 will now be chosen so as to obtain the highest lower bound and the lowest upper bound from (3.18). It can be shown that the right-hand side of (3.18) is a monotonically increasing function of μ_0. Accordingly the best choice of μ_0 which complies with (2.13a) is

$$\mu_0=\mu_1, \quad (3.19)$$

where μ_1 is the smallest of the μ_t. The best choice that complies with (2.13b) is

$$\mu_0=\mu_m, \quad (3.20)$$

where μ_m is the largest of μ_t. Accordingly,

$$\mu_1{}^*=\mu_1+A_1/(1-\alpha_1A_1), \quad (3.21a)$$

where

$$\alpha_1=(3\mu_1)^{-1}, \quad (3.21b)$$

$$A_1=\sum_{t=2}^m v_t/[(\mu_t-\mu_1)^{-1}+\alpha_1], \quad (3.21c)$$

and

$$\mu_m{}^*=\mu_m+A_m/(1-\alpha_mA_m), \quad (3.22a)$$

where

$$\alpha_m=(3\mu_m)^{-1}, \quad (3.22b)$$

$$A_m=\sum_{t=1}^{m-1}v_t/[(\mu_t-\mu_m)+\alpha_m], \quad (3.22c)$$

and μ^* satisfies the inequality

$$\mu_1{}^*<\mu^*<\mu_m{}^*. \quad (3.23)$$

Finally it is interesting to note that for $\mu_0\to0$ and $\mu_0\to\infty$, which are the worst possible choices for μ_0, the bounds (3.18) reduce to

$$\sum_{t=1}^m v_t\mu_t>\mu^*>[\sum_{t=1}^m(v_t/\mu_t)]^{-1}. \quad (3.24)$$

These bounds have been first derived by Wiener,[2] and later on by a variational approach by Brown[11] (see also reference 6).

[11] W. F. Brown (private communication).

IV. APPLICATION TO TWO-PHASE MATERIALS

An important question is whether the latitude between the bounds derived above is an inherent property of the model studied or is it due, at least in part, to some limitations of the method used. For the general case this question cannot be answered at present. Fortunately, however, this problem can be solved for the important case of two-phase materials.

Consider a homogeneous body of permeability μ. On the surface of the body a potential ψ_0 is prescribed which creates within the body a homogeneous field H_0. A sphere of radius r_b is now replaced by a composite sphere consisting of an inner part of radius r_a and permeability μ_a and a concentric shell with permeability μ_b. Under what conditions then will there be no change in the total energy stored in the body? It is clear that if the field outside of r_b remains unchanged, then there will be no change in energy due to the replacement.

In a spherical coordinate system (r,ϑ,φ) whose axis $\vartheta=0$ coincides with $\mathbf{H}_0$ and whose center coincides with the center of the sphere considered, the potential before the change is

$$\psi=-H_0 r\cos\vartheta. \tag{4.1a}$$

This is therefore taken to be the potential after the change for

$$r\geqslant r_b. \tag{4.1b}$$

From Laplace's equation it follows that

$$\psi=(C_1 r+C_2 r^{-2})\cos\vartheta \tag{4.2a}$$

for

$$r_b\geqslant r\geqslant r_a, \tag{4.2b}$$

and that

$$\psi=C_3 r\cos\vartheta \tag{4.3a}$$

for

$$r_a\geqslant r. \tag{4.3b}$$

The boundary conditions at r_b and r_a require that

$$H_0 r_b-C_1 r_b-C_2 r_b^{-2}=0, \tag{4.4a}$$

$$\mu H_0-\mu_b C_1+2\mu_b C_2 r_b^{-3}=0, \tag{4.4b}$$

$$C_1 r_a+C_2 r_a^{-2}-C_3 r_a=0, \tag{4.4c}$$

$$\mu_b C_1-2\mu_b C_2 r_a^{-3}-\mu_a C_3=0. \tag{4.4d}$$

In order that (4.4) be self-consistent the determinant of the coefficients of H_0, C_1, C_2, and C_3 must vanish, i.e.,

$$r_a^{-3}\begin{vmatrix} -1 & 1 & (r_a/r_b)^3 & 0 \\ -\mu & \mu_b & -2\mu_b(r_a/r_b)^3 & 0 \\ 0 & 1 & 1 & -1 \\ 0 & \mu_b & -2\mu_b & -\mu_a \end{vmatrix}=0, \tag{4.5}$$

which reduces to

$$\mu=\mu_b+\frac{(r_a/r_b)^3}{\dfrac{1}{\mu_a-\mu_b}+\dfrac{1-(r_a/r_b)^3}{3\mu_b}}. \tag{4.6a}$$

From the above considerations it follows that the effective permeability of the body considered is not influenced by the change made, provided (4.6a) holds. Thus if this change is repeated indefinitely until all the original material is replaced by composite spheres, which may diminish to infinitesimal size, but all with the same r_a/r_b, a macroscopically isotropic and homogeneous material can be constructed whose effective permeability is exactly given by (4.6a). For this particular material

$$v_a=(r_a/r_b)^3, \tag{4.6b}$$

$$v_b=1-(r_a/r_b)^3. \tag{4.6c}$$

Now for the case $m=2$, the bounds (3.21) and (3.22) derived in Sec. III reduce to

$$\mu_1{}^*=\mu_1+\frac{v_2}{\dfrac{1}{\mu_2-\mu_1}+\dfrac{v_1}{3\mu_1}} \tag{4.7}$$

and

$$\mu_2{}^*=\mu_2+\frac{v_1}{\dfrac{1}{\mu_1-\mu_2}+\dfrac{v_2}{3\mu_2}}. \tag{4.8}$$

For $\mu_{1,2}=\mu_{a,b}$, $v_{1,2}=v_{a,b}$ (4.6) is equal to (4.7).

For $\mu_{1,2}=\mu_{b,a}$, $v_{1,2}=v_{b,a}$ (4.6) is equal to (4.8).

It follows that (4.7) and (4.8) are the best possible bounds for the magnetic permeability of macroscopically homogeneous and isotropic two-phase materials which can be derived in terms of the phase volume fractions and phase permeabilities.[12] To improve the bounds additional information in the statistics of the spatial distribution of the phases is needed.

V. DISCUSSION

Extensive experimental studies of the effective properties of multiphase materials are available.[2–6] However, most of these have been restricted to two-phase materials. For this case it has also been shown above that the bounds obtained are the most restrictive ones. Comparison with experimental results will hence be restricted to two-phase materials only. Perhaps the most extensive experimental work for this case is that of DeLoor[3] who studied the dielectric constant of mixtures and compared his results with different theoretical formulas. He concluded that "no mixture 'law' proposed up to the present gives an accurate description of the dielectric behavior of a heterogeneous mixture, and for each individual case it is still necessary to determine the particular parameters." However, he finds that the results always fall between two limits, which interest-

[12] A similar conclusion was reached in a study of the effective elastic moduli of multiphase materials. Z. Hashin and S. Shtrikman (to be published).

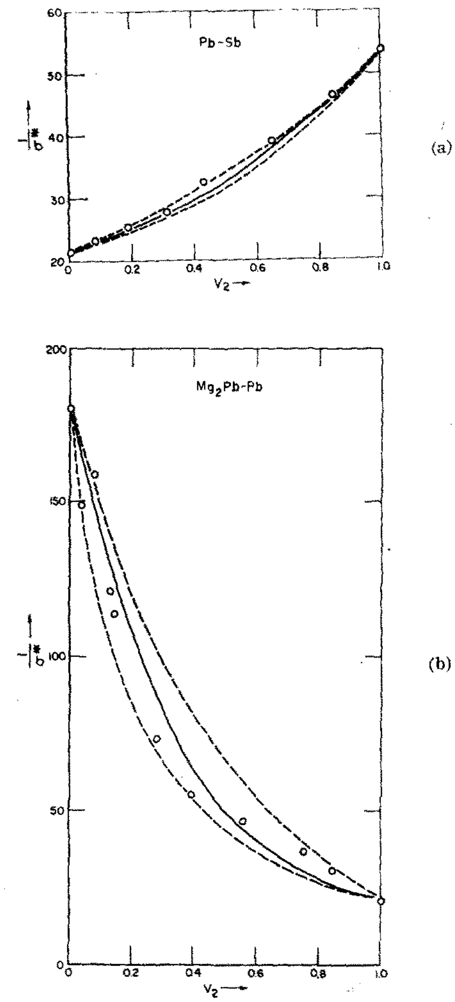

FIG. 1 (a) and (b). Resistivity $1/\sigma^*$ of two-phase alloys as a function of the concentration v_2 of one of the phases. Experimental points are taken from Landauer. The solid line is computed from Eq. (5) replacing the permeability μ by the conductivity σ. The broken line describes the bounds given by (4.7) and (4.8), where again μ has been replaced by σ.

ingly enough coincide with (4.7) and (4.8). (These expressions coincide with the so-called Maxwell formula.[7])

A relation which has been extensively used in the literature is that given by Böttcher[2]

$$v_1\frac{\mu_1-\mu^*}{\mu_1+2\mu^*}=v_2\frac{\mu_2-\mu^*}{\mu_2+2\mu^*}. \tag{5.1}$$

This has been found by Böttcher and by Landauer[13] to be a reasonably good description of the experimental data which they consider. It should be noted that μ^* given by (5.1) always lies between the bounds (4.7) and (4.8) so that these results are not in disagreement with the present bounds. Figures 1(a) and 1(b) give, as an example, two of the cases considered by Landauer, together with two of the present bounds. There is a significant discrepancy between (5.1) and the experimental values shown in Fig. (1b). Again, however, and as observed by DeLoor,[3] the experimental results lie between the theoretical bounds (4.7) and (4.8).

[13] R. Landauer, J. Appl. Phys. **23**, 779 (1952).

In conclusion, one can therefore say that the bounds (4.7) and (4.8) give a reasonably good description of the experimentally observed effective constants of two-phase materials. The indeterminacy of μ^* is an inherent property of the physical situation, resulting from the fact that generally nothing is known about the spatial distribution of the phases except that the material is macroscopically homogeneous and isotropic. This indeterminacy cannot be reduced without additional physical information.

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APPENDIX. PROOF OF VARIATIONAL PRINCIPLES

The variation ΔU_T of U_T with respect to a change $\delta\mathbf{T}$ in $\mathbf{T}$ is composed only of first- and second-order terms in $\delta\mathbf{T}$, i.e.,

$$\Delta U_T=\delta U_T+\delta^2 U_T. \tag{A1}$$

These will be treated separately. From (2.9)

$$\delta U_T=\frac{1}{8\pi}\int_{(V)}\left\{2\left(-\frac{\mathbf{T}}{\mu-\mu_0}+\mathbf{H}_0+\mathbf{H}'\right)\cdot\delta\mathbf{T}-\delta\mathbf{T}\cdot\mathbf{H}'+\mathbf{T}\cdot\delta\mathbf{H}'\right\}dv. \tag{A2}$$

The expression in the parentheses multiplying $\delta\mathbf{T}$ vanishes when (2.12) is true, as follows from the definition (2.8). Expression (A2) therefore reduces to

$$\delta U_T=\frac{1}{8\pi}\int_{(V)}(-\delta\mathbf{T}\cdot\mathbf{H}'+\mathbf{T}\cdot\delta\mathbf{H}')dv. \tag{A3}$$

The subsidiary condition (2.10) can be integrated in the form

$$\mu_0\mathbf{H}'+\mathbf{T}=\mathbf{C}, \tag{A4}$$

where

$$\operatorname{div}\mathbf{C}=0. \tag{A5}$$

Substituting for $\mathbf{T}$ and $\delta\mathbf{T}$ from (A4) into (A3) results in

$$\delta U_T=\frac{1}{8\pi}\int_{(V)}(-\delta\mathbf{C}\cdot\mathbf{H}'+\mathbf{C}\cdot\delta\mathbf{H}')dv, \tag{A6}$$

which vanishes because of (2.15) when taking into account (A5), (2.11) and the relation

$$H'=-\operatorname{grad}\psi'. \tag{A7}$$

Consider now the second term $\delta^2 U_T$ in ΔU_T

$$\delta^2 U_T=\frac{1}{8\pi}\int_{(V)}\left(\frac{(\delta\mathbf{T})^2}{\mu-\mu_0}+\delta\mathbf{T}\cdot\delta\mathbf{H}'\right)dv. \tag{A8}$$

Substituting (A4) into (A8) and using

$$\int_{(V)}(\delta\mathbf{C}\cdot\delta\mathbf{H}')dv=0, \tag{A9}$$

which holds for the same reasons for which (A6) vanishes, it is found that

$$\delta^2 U_T=\frac{1}{8\pi}\int_{(V)}\left\{-\frac{(\delta\mathbf{C})^2}{\mu-\mu_0}-\mu_0(\delta\mathbf{H}')^2\right\}dv. \tag{A10}$$

Substituting (A10) in (A1) and remembering that $\delta U_T=0$, (2.13a) follows. To prove (2.13b), the relation

$$\int_{(V)}(\delta\mathbf{T})^2 dv=\int_{(V)}\{\mu_0^2(\delta\mathbf{H}')^2+(\delta\mathbf{T})^2\}dv, \tag{A11}$$

which follows from (A4) and (A9), is used to eliminate $(\delta H')^2$ from (A3), which then assumes the form

$$\delta^2 U_T=\frac{1}{8\pi}\int_{(V)}\left\{-\left(\frac{1}{\mu-\mu_0}+\frac{1}{\mu_0}\right)(\delta\mathbf{T})^2+\frac{1}{\mu_0}(\delta\mathbf{C})^2\right\}dv. \tag{A12}$$

For $\mu<\mu_0$, $\delta^2 U_T$ is positive so that with $\delta U_T=0$, (2.13b) follows from (A1).

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A Rapid X-Ray Method for the Determination of Substructure Characteristics

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An x-ray diffraction method is described by which the substructure characteristics and dislocation densities of crystalline materials can be rapidly determined. The method is based on spatial tracing of reflection images and their extrapolation onto the specimen surface using the method of least squares.

INTRODUCTION

FOR the study of the mechanical properties of metals and alloys it is frequently desirable to obtain speedily some quantitative measurements of the substructure characteristics. Previously, one of the authors (S.W.) has described a method by which the subgrain size, the disorientation angle between adjacent subgrains, and the lattice misalignment within the subgrains can be accurately determined for single crystals[1] as well as for polycrystalline materials.[2,3] While this method is capable of a very high resolution, it is time-consuming and leads to detailed structural information which may not be essential for a particular type of investigation.

It is the aim of this paper to present an x-ray method based on the principle of image tracing by which the parameters describing the substructure of a polycrystalline specimen as well as single crystals can be quantitatively determined with considerable speed. This method may be regarded as an analytical extension of the Berg-Barrett[4,5] and Schulz[6] x-ray topography methods.

This method has the added advantage that the requirements for instrumentation are reduced to a minimum and that its application, therefore, is within the reach of any laboratory having standard diffraction equipment. Although the image-tracing technique for the study of substructure has been previously described,[2,3,7] this paper concerns itself with the extension of the quantitative aspects of the method.

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